

# INTRODUCTION

The objective of this homework is to apply fundamental Signals and Systems concepts to a real audio signal. A sequence of musical notes is recorded using a smartphone and analyzed in both time and frequency domains using MATLAB. The recorded signal is examined to estimate the fundamental frequencies of the musical notes, and a digital band-pass filter is designed to isolate the **Re (D) note** as the target component. In addition, a simple MATLAB graphical user interface (GUI) is implemented[5] to visualize the signals and play the original and filtered audio.

## DATA COLLECTION

The audio signal used in this study was obtained by recording a sequence of musical notes played on a Casio CTX-800 digital keyboard. The notes were selected from the C4–C5 octave range, and a total of eight distinct notes were played sequentially. The recording was captured using a Samsung Galaxy M52 smartphone with the device’s built-in voice recording application.

The total duration of the recording was approximately 17 seconds. A waiting interval of about one second was left between consecutive notes to clearly separate them in the time domain. The recording was performed under home conditions in a relatively quiet indoor environment in order to minimize background noise and external disturbances.

After the recording process, the audio file was transferred to a personal computer. All subsequent signal processing, analysis, and visualization tasks were carried out using MATLAB.

## METHOD

The recorded audio file was first imported into MATLAB using the **audioread** function, where the signal was represented as a discrete-time sequence  $x[n]$  together with its sampling frequency  $F_s$ . If the recording contained more than one channel (stereo), the channels were averaged to obtain a single mono signal. The total number of samples was denoted by  $N$ , and the total duration of the signal was calculated as

$$T = \frac{N}{F_s}$$

As an initial step, the recorded signal was played once to obtain a qualitative impression of the note sequence.

The original signal was first analyzed in the time domain. In **Figure 1**, the complete waveform of the recorded audio signal is shown. Based on this plot, approximate time intervals corresponding to each musical note were identified. Using these intervals, individual note segments were extracted and examined separately in the time domain. The time-domain

waveforms of these individual note segments are presented in **Figure 2**, where their amplitude characteristics and temporal separation can be observed.

For frequency-domain analysis, the Fourier Transform of each note segment was computed using the Fast Fourier Transform (FFT)[4]. The magnitude spectra of the individual notes are shown in **Figure 3**, where the dominant peaks corresponding to the fundamental frequencies of the notes were identified. In addition, the phase spectra of the same note segments were examined and are presented in **Figure 4**.

After identifying the fundamental frequency of the Re (D) note, a band-pass digital filter was designed to isolate this note from the full recording. Let  $f_0$  denote the estimated fundamental frequency of the Re note. The passband of the filter was selected as

$$f_L = f_0 - \Delta f, \quad f_H = f_0 + \Delta f,$$

where  $\Delta f$  represents a margin chosen to include the main spectral lobe of the target note while attenuating neighboring notes. The corresponding normalized digital frequencies were calculated as

$$\omega_L = \frac{2f_L}{F_s}, \omega_H = \frac{2f_H}{F_s}.$$

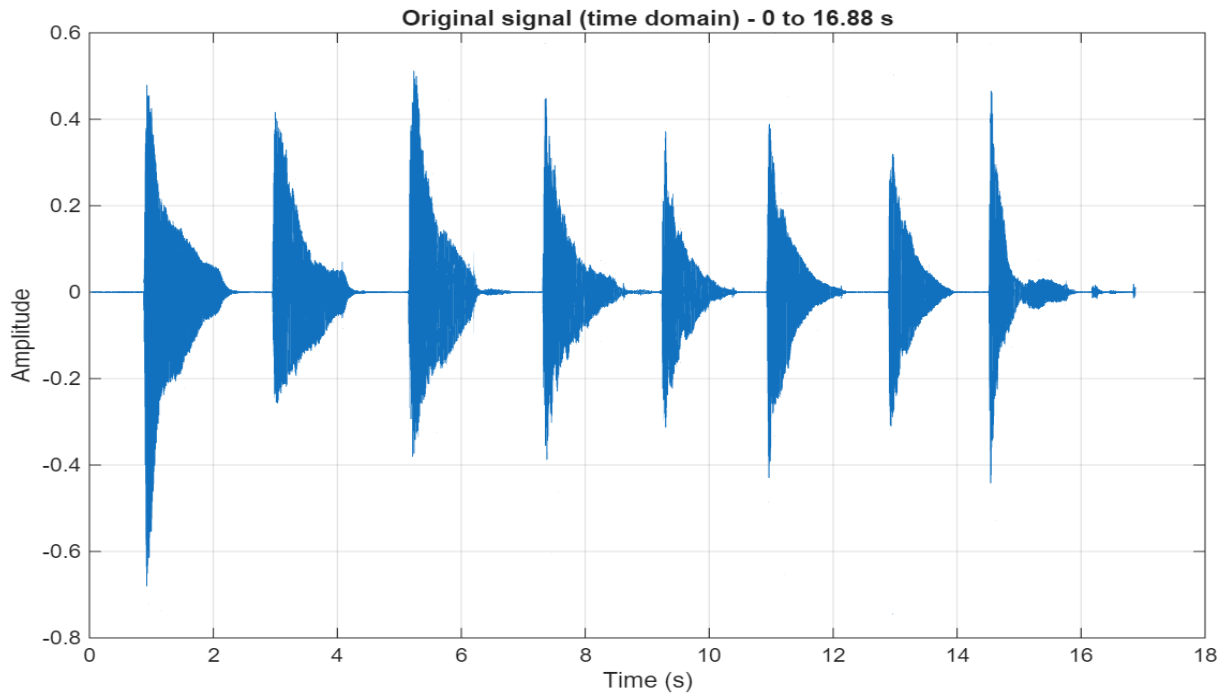
Using these values, a band-pass filter was designed. The magnitude and phase responses of the designed filter in the frequency domain are shown in **Figure 5**.

The designed band-pass filter was applied to the entire recorded signal. The time-domain comparison of the original and filtered signals is presented in **Figure 6**, where the target Re note is clearly emphasized while other notes are significantly attenuated. However, it is observed that residual components from the first and third notes are still present after filtering. This limitation is mainly due to non-ideal recording conditions in a home environment, finite note durations, limited frequency resolution, and spectral overlap caused by harmonic components that fall within the selected passband. The frequency-domain comparison of the original and filtered signals is shown in **Figure 7**, which demonstrates that the energy around the fundamental frequency of the Re note is preserved, whereas most of the remaining frequency components are effectively suppressed.

A MATLAB graphical user interface (GUI) was developed to provide an interactive environment for audio analysis and filtering. The design objectives of the GUI were defined at the beginning of the implementation. Buttons were created to load the audio file, play the original signal, apply the selected filter, and play the filtered signal. To inform the user about the loaded audio file, the sampling frequency  $F_s$  and the total signal duration were displayed within the interface.

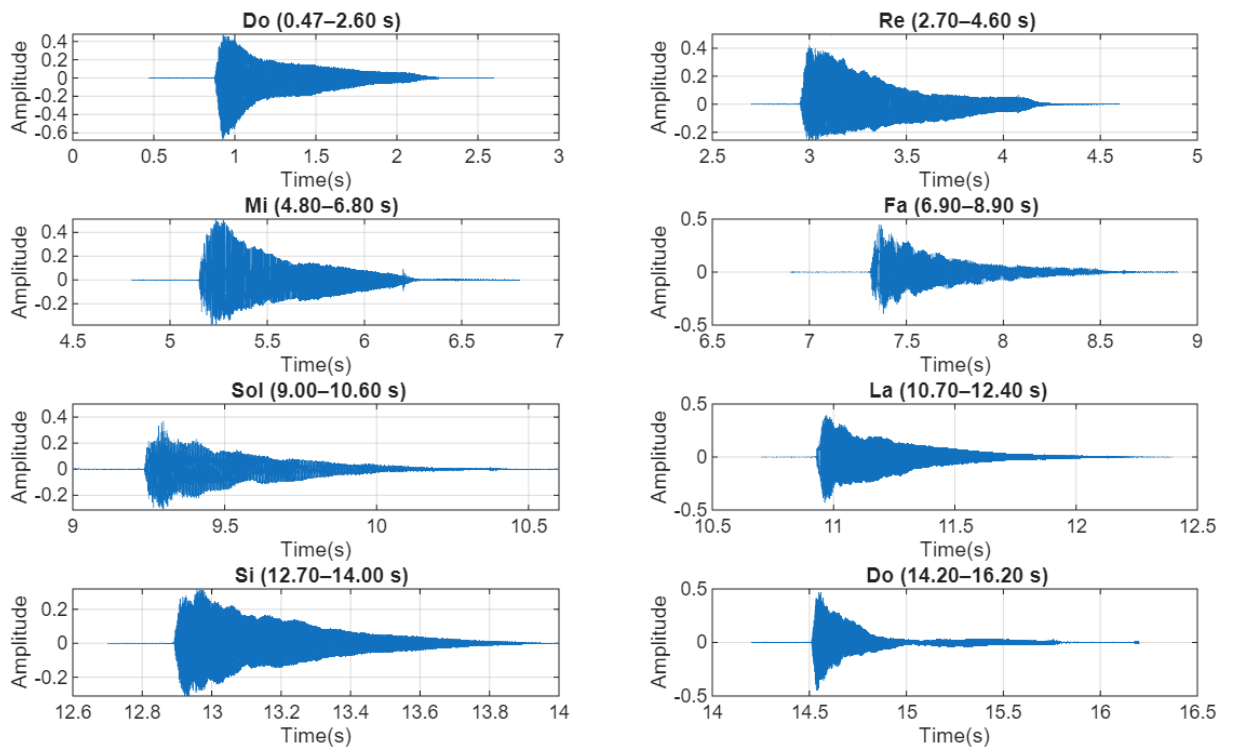
For visual comparison, time-domain and frequency-domain plots of the original and filtered signals were placed in an upper-lower layout, allowing a direct and clear comparison of filtering effects. In addition to standard low-pass, high-pass, and band-pass filter options, a custom-designed band-pass filter specifically tuned to the target Re note was included as a selectable option. This GUI structure enables both visual and auditory evaluation of the filtering process and improves the usability and clarity of the analysis.

# RESULTS



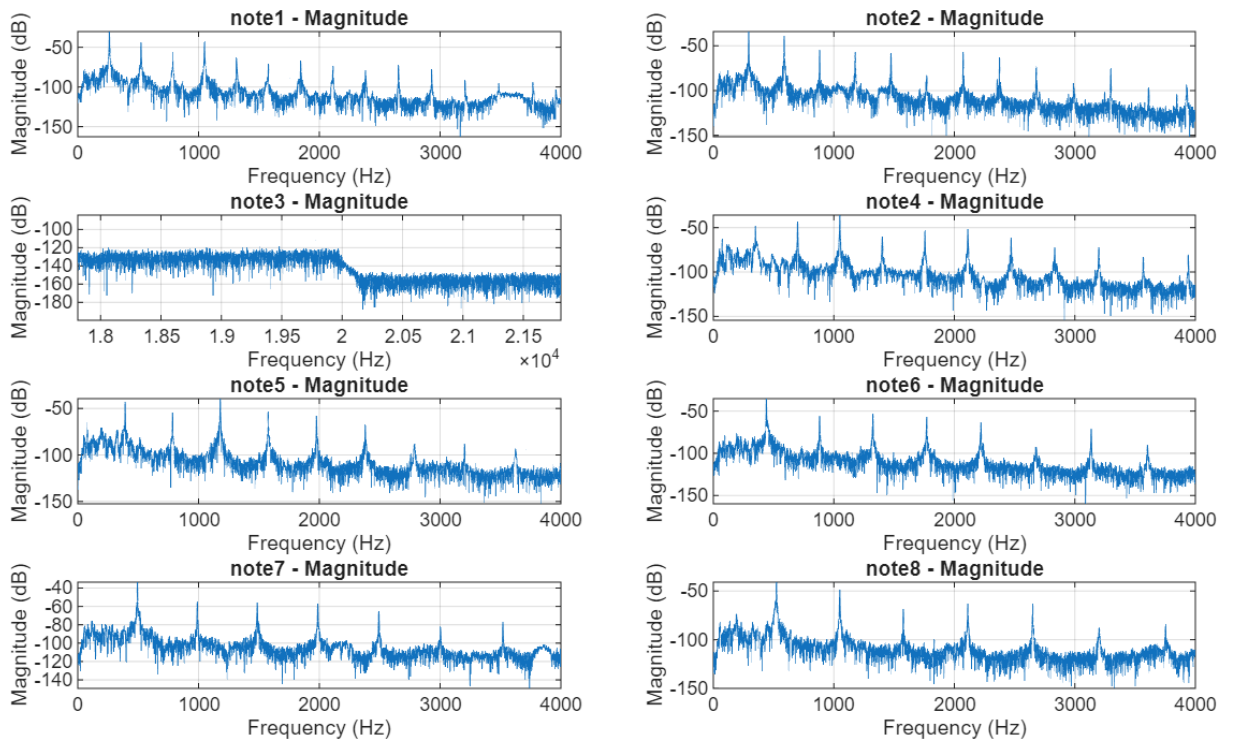
**Figure 1.** Time-domain waveform of the original audio signal recorded from the digital keyboard, showing the sequence of eight musical notes.

## Short time segments of each note (time domain)



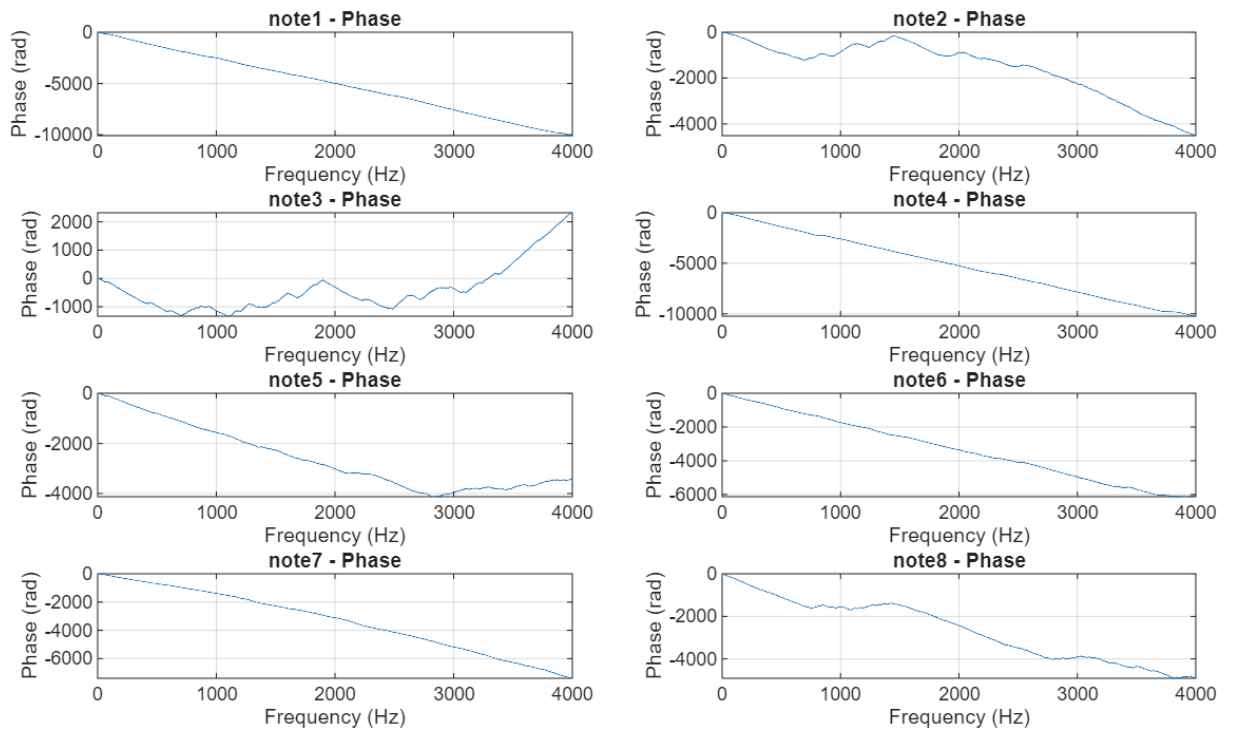
**Figure 2.** Time-domain waveforms of the individual musical note segments extracted from the original recording, showing the amplitude variations of each note within its selected time interval.

### Magnitude Spectra of Segmented Notes (Positive Frequencies)

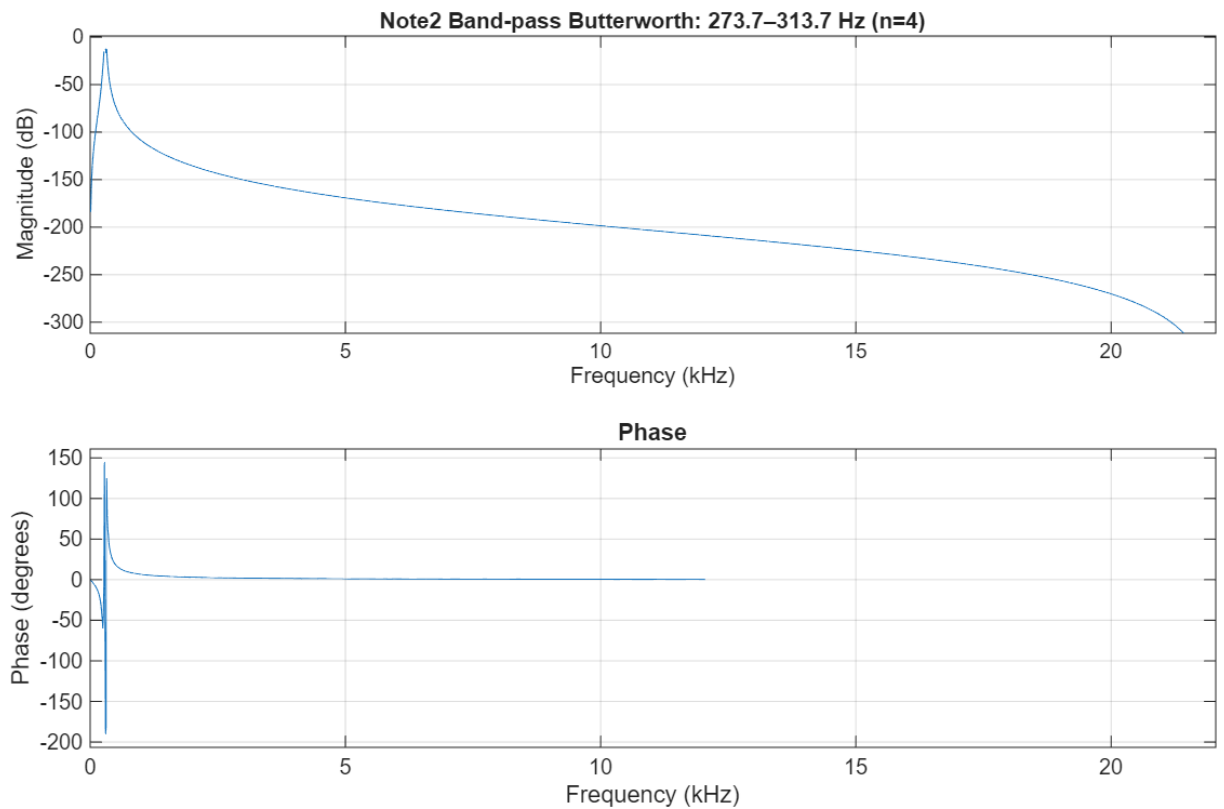


**Figure 3.** Magnitude spectra of the individual musical note segments, illustrating the fundamental frequencies and similar harmonic structures resulting from the same sound source.

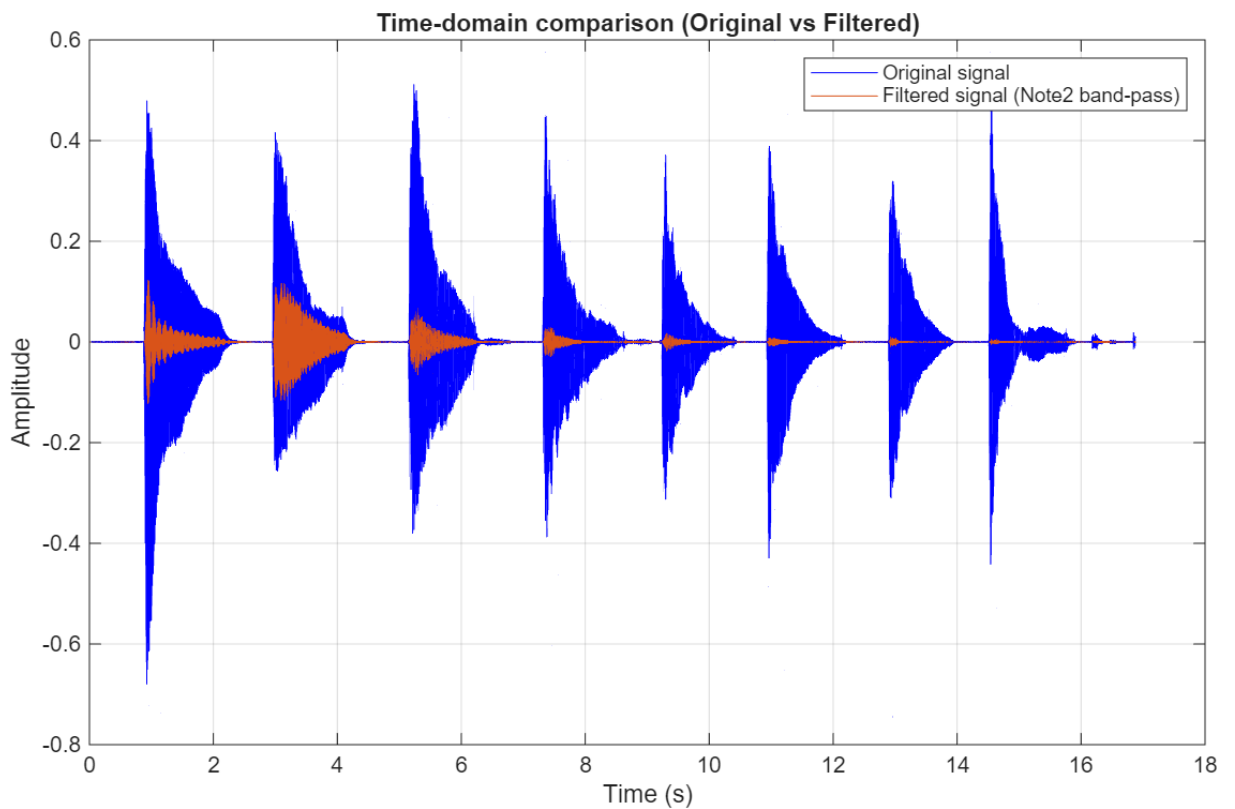
### Phase Spectra of Segmented Notes (Positive Frequencies)



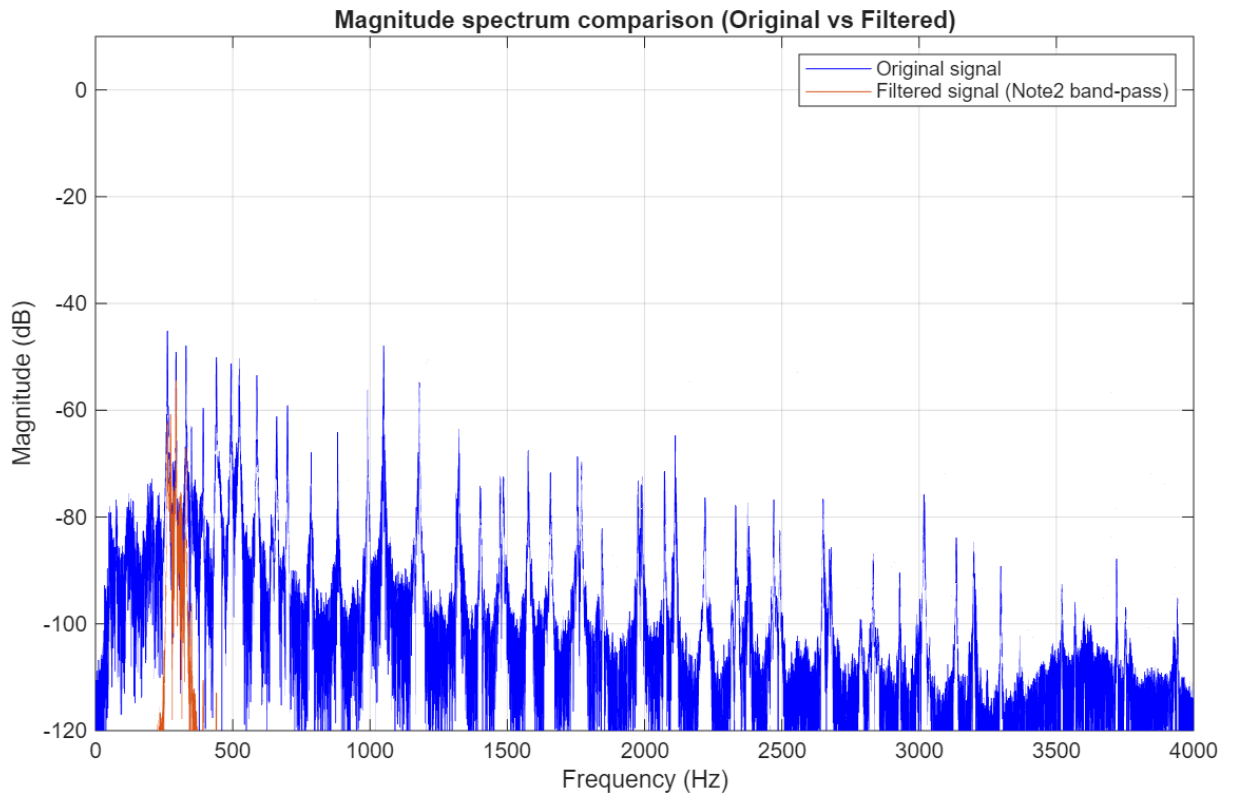
**Figure 4.** Phase spectra of the segmented musical notes computed over the positive frequency range, illustrating the phase characteristics of each note.



**Figure 5.** Magnitude and phase responses of the designed Butterworth band-pass filter used to isolate the Re (D) note, with a passband of 273.7–313.7 Hz and filter order  $n = 4$ .



**Figure 6**



**Figure 7**

## DISCUSSION AND CONCLUSION

In this study, fundamental Signals and Systems concepts were successfully applied to a real audio signal recorded under non-ideal conditions. Time-domain analysis allowed the identification and segmentation of individual musical notes, while frequency-domain analysis enabled the estimation of their fundamental frequencies and harmonic structures. The similarity observed among the magnitude spectra of different notes is primarily due to the use of the same instrument, which results in similar spectral envelopes and harmonic content.

A band-pass Butterworth filter was designed[3G] to isolate the Re (D) note from the recorded signal. Although the filter effectively preserved the target note and significantly attenuated other components, complete suppression of neighboring notes was not achieved. This limitation can be attributed to factors such as finite note durations, limited frequency resolution, spectral overlap between adjacent notes, and the presence of harmonics within the selected passband. Additionally, background noise and recording conditions in a home environment contributed to residual spectral components after filtering.

Overall, this homework demonstrated the practical challenges of applying theoretical signal processing techniques to real-world audio data. The results highlight the importance of careful filter design, appropriate parameter selection, and an understanding of spectral characteristics when working with non-ideal signals. The implemented MATLAB GUI provided an effective platform for visualizing and evaluating the analysis and filtering process.

# ABSTRACT

In this homework, a real audio signal consisting of a sequence of musical notes was recorded and analyzed using fundamental Signals and Systems concepts. The recorded signal was examined in both time and frequency domains to identify note characteristics and estimate fundamental frequencies. A digital band-pass Butterworth filter was designed to isolate the Re (D) note, and the filtering performance was evaluated through visual and auditory comparisons. In addition, a MATLAB graphical user interface (GUI) was developed to facilitate signal loading, visualization, filtering, and playback. While the overall system design, signal analysis methodology, filter parameter selection, and GUI structure were carried out by the student, parts of the MATLAB code implementation were generated with the assistance of artificial intelligence tools. This study highlights both the practical challenges of real-world audio processing and the effective integration of theoretical concepts with computational tools.

## References:

- [1] A. V. Oppenheim, A. S. Willsky, and S. H. Nawab, Signals and Systems, 2nd ed., Prentice Hall, Upper Saddle River, NJ, USA, 1997.
- [2] MathWorks, MATLAB Signal Processing Toolbox Documentation, <https://www.mathworks.com/help/signal/>
- [3] MathWorks, Digital Filter Design (Butterworth Filters), <https://www.mathworks.com/help/signal/ref/butter.html>
- [4] MathWorks, Fast Fourier Transform (FFT), <https://www.mathworks.com/help/matlab/ref/fft.html>
- [5] MathWorks, MATLAB App Designer Documentation, <https://www.mathworks.com/help/matlab/app-designer.html>